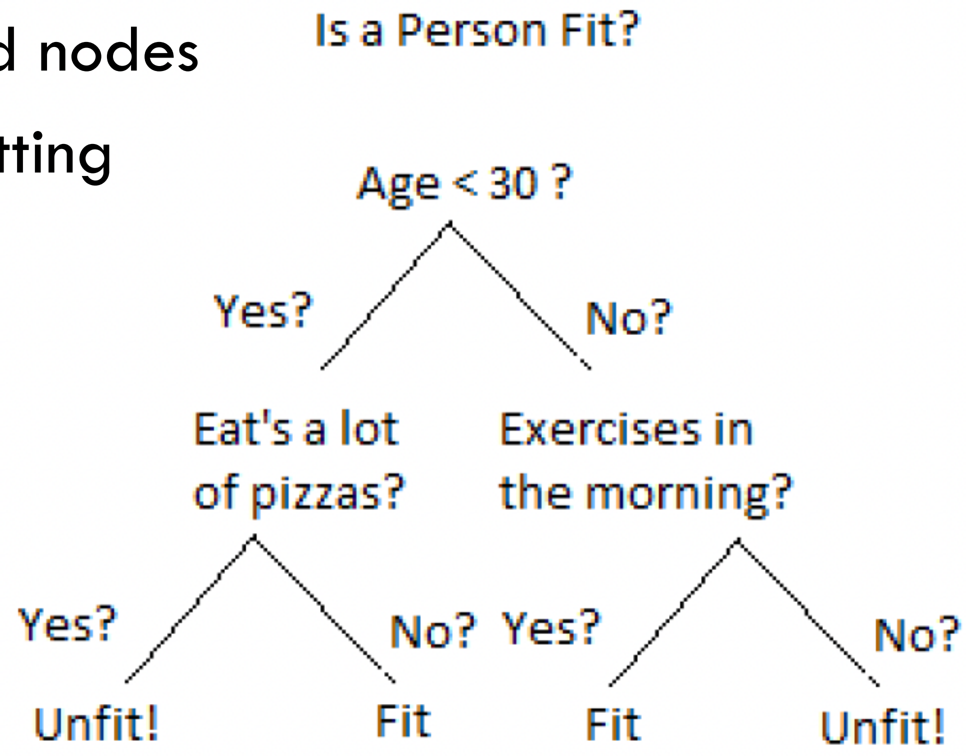
DECISION TREE ALGORITHM

Decision Tree Algorithm

- Root node
- Parent, child nodes
- Branch, splitting
- Leaf nodes



Decision Tree Algorithm

- Training dataset

Day	Outlook	Temp	Humidity	Wind	Tennis?
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
3	Overcast	Hot	High	Weak	Yes
4	Rain	Mild	High	Weak	Yes
5	Rain	Cool	Normal	Weak	Yes
6	Rain	Cool	Normal	Strong	No
7	Overcast	Cool	Normal	Strong	Yes
8	Sunny	Mild	High	Weak	No
9	Sunny	Cool	Normal	Weak	Yes
10	Rain	Mild	Normal	Weak	Yes
11	Sunny	Mild	Normal	Strong	Yes
12	Overcast	Mild	High	Strong	Yes
13	Overcast	Hot	Normal	Weak	Yes
14	Rain	Mild	High	Strong	No

Decision Tree Algorithm

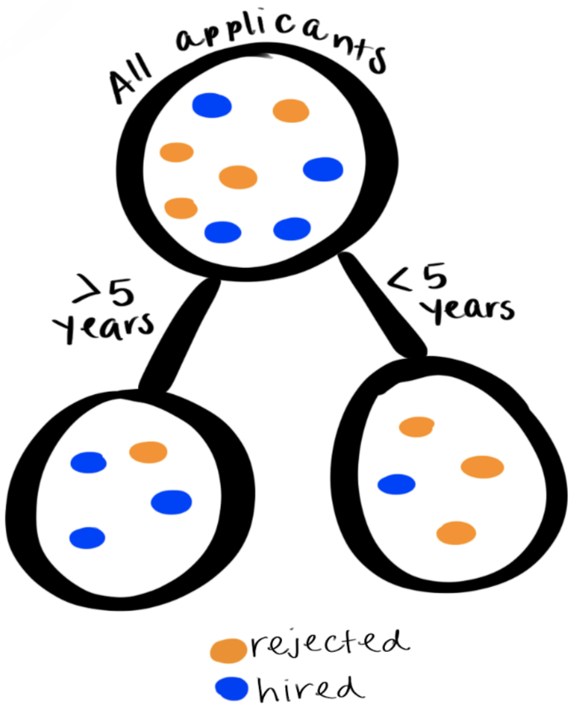
- How can we build a decision tree given a data set?

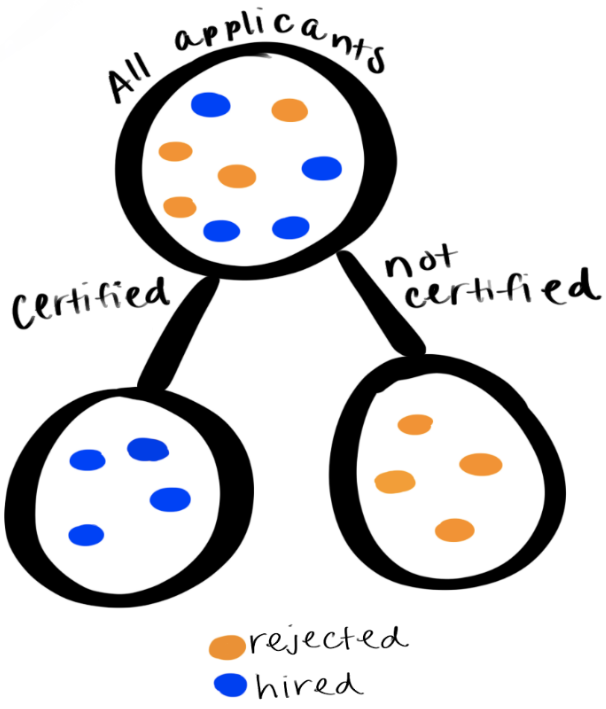
Decision Tree Algorithm

- Decision trees look at one feature/input variable at a time/node
- We will make the best choice at each node
  - Identify the best feature/input variable for the each node

Decision Tree Algorithm

- Identify the best feature/input variable for root node
  - Best split: results of each branch should be as homogeneous (or pure) as possible





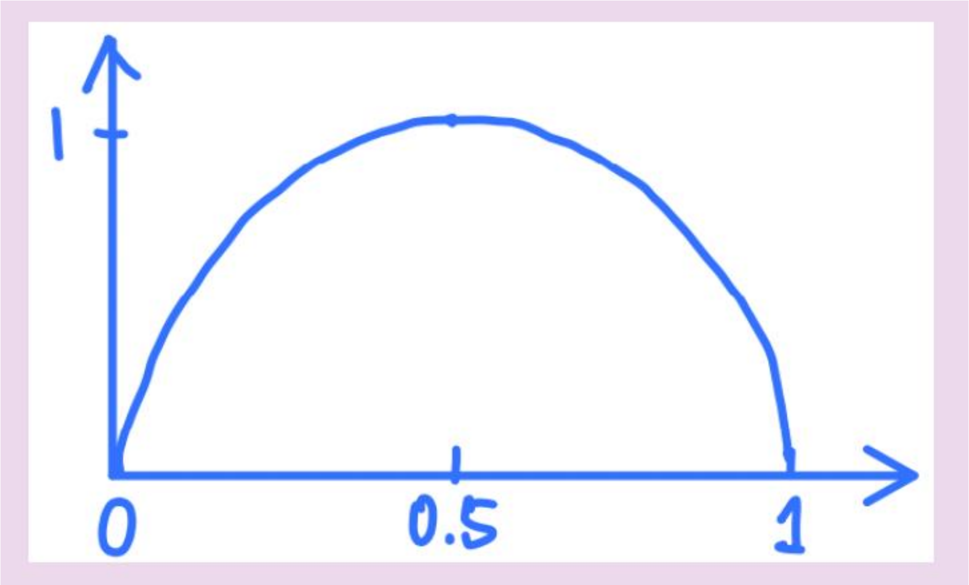
Decision Tree Algorithm

- Identify the best feature/attribute for root node
  - Best split: results of each branch should be as homogeneous (or pure) as possible
  - a feature that reduces impurity as much as possible

  - How do we measure the impurity in a set of examples
    - Entropy from information theory
    - Alternatively, use Gini

Decision Tree Algorithm

- Entropy for a distribution over two outcomes



Decision Tree Algorithm

- Quantifying the information content of a feature
  - Impurity Reduction or Information gain or entropy reduction

$$\text{InfoGain} = I_{\text{before}} - I_{\text{after}}$$

Decision Tree Algorithm

- Entropy of the examples before we select a feature for the root node

$$H_{\text{before}} = -\left(\frac{9}{14}\log_2\left(\frac{9}{14}\right) + \frac{5}{14}\log_2\left(\frac{5}{14}\right)\right)$$
$$\approx 0.94$$

Day	Outlook	Temp	Humidity	Wind	Tennis?
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
3	Overcast	Hot	High	Weak	Yes
4	Rain	Mild	High	Weak	Yes
5	Rain	Cool	Normal	Weak	Yes
6	Rain	Cool	Normal	Strong	No
7	Overcast	Cool	Normal	Strong	Yes
8	Sunny	Mild	High	Weak	No
9	Sunny	Cool	Normal	Weak	Yes
10	Rain	Mild	Normal	Weak	Yes
11	Sunny	Mild	Normal	Strong	Yes
12	Overcast	Mild	High	Strong	Yes
13	Overcast	Hot	Normal	Weak	Yes
14	Rain	Mild	High	Strong	No

Decision Tree Algorithm

- Information gain if we select Outlook for the root node

$$\text{Outlook} = \begin{cases} \text{Sunny} & 2+ \quad 3- \quad 5 \text{ total} \\ \text{Overcast} & 4+ \quad 0- \quad 4 \text{ total} \\ \text{Rain} & 3+ \quad 2- \quad 5 \text{ total} \end{cases}$$

$$\text{Gain(Outlook)} = 0.94 - \left(\frac{5}{14} \cdot I\left(\frac{2}{5}, \frac{3}{5}\right) + \frac{4}{14} \cdot I\left(\frac{4}{4}, \frac{0}{4}\right) + \frac{5}{14} \cdot I\left(\frac{3}{5}, \frac{2}{5}\right)\right)$$
$$= 0.247$$

Day	Outlook	Temp	Humidity	Wind	Tennis?
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
3	Overcast	Hot	High	Weak	Yes
4	Rain	Mild	High	Weak	Yes
5	Rain	Cool	Normal	Weak	Yes
6	Rain	Cool	Normal	Strong	No
7	Overcast	Cool	Normal	Strong	Yes
8	Sunny	Mild	High	Weak	No
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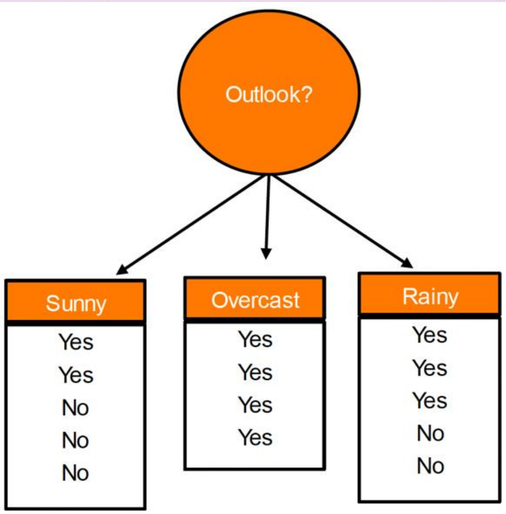

Decision Tree Algorithm

- Information gain if we select Humidity for the root node

$$\text{Humidity} = \begin{cases} \text{Normal} & 6+ \quad 1- \quad 7 \text{ total} \\ \text{High} & 3+ \quad 4- \quad 7 \text{ total} \end{cases}$$

$$\text{Gain(Humidity)} = 0.94 - \left(\frac{7}{14} \cdot I\left(\frac{6}{7}, \frac{1}{7}\right) + \frac{7}{14} \cdot I\left(\frac{3}{7}, \frac{4}{7}\right) \right)$$

$$= 0.151$$

Day	Outlook	Temp	Humidity	Wind	Tennis?
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
3	Overcast	Hot	High	Weak	Yes
4	Rain	Mild	High	Weak	Yes
5	Rain	Cool	Normal	Weak	Yes
6	Rain	Cool	Normal	Strong	No
7	Overcast	Cool	Normal	Strong	Yes
8	Sunny	Mild	High	Weak	No
9	Sunny	Cool	Normal	Weak	Yes
10	Rain	Mild	Normal	Weak	Yes
11	Sunny	Mild	Normal	Strong	Yes
12	Overcast	Mild	High	Strong	Yes
13	Overcast	Hot	Normal	Weak	Yes
14	Rain	Mild	High	Strong	No

Decision Tree Algorithm

- Outlook has the greatest information gain

$\text{Gain}(\textbf{Outlook}) = \mathbf{0.247}$ $\quad$ $\text{Gain}(\text{Humidity}) = 0.151$

$\text{Gain}(\text{Temp}) = 0.029$ $\quad$ $\text{Gain}(\text{Wind}) = 0.048$

Day	Outlook	Temp	Humidity	Wind	Tennis?
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
3	Overcast	Hot	High	Weak	Yes
4	Rain	Mild	High	Weak	Yes
5	Rain	Cool	Normal	Weak	Yes
6	Rain	Cool	Normal	Strong	No
7	Overcast	Cool	Normal	Strong	Yes
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12	Overcast	Mild	High	Strong	Yes
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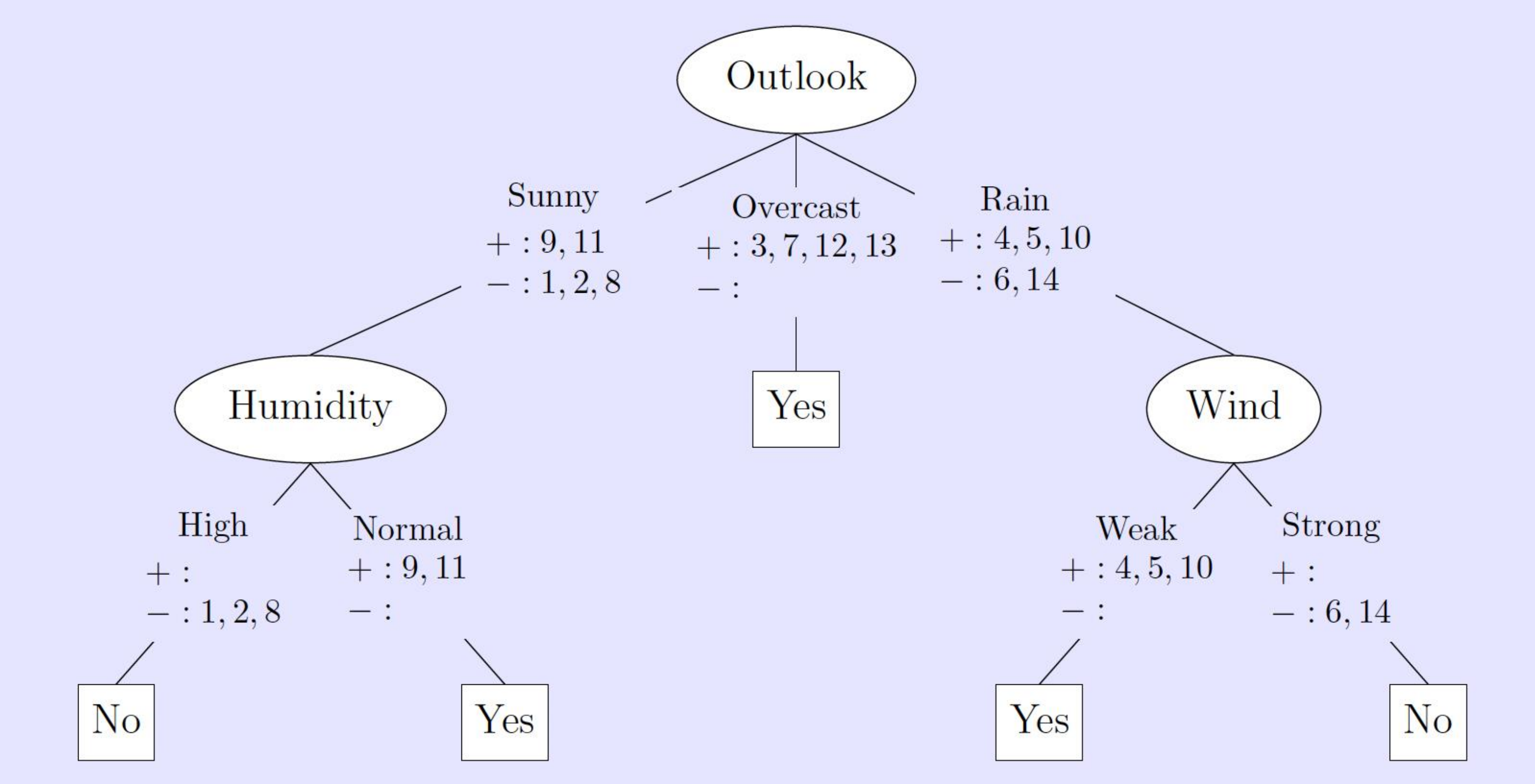
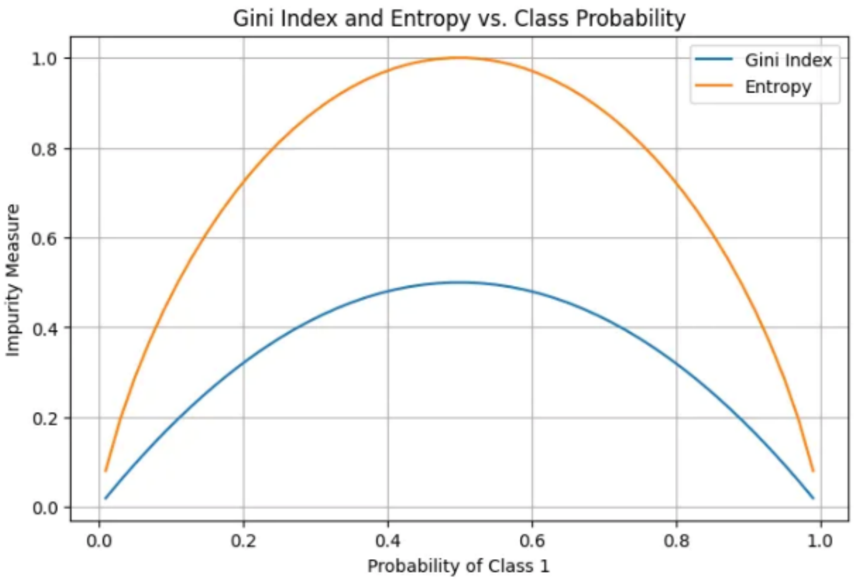
Decision Tree Algorithm

- Complete Decision Tree



Gini Impurity to Build Decision Trees

$$Gini(D) = 1 - \sum_{i=1}^{k} p_i^2$$



Thank You!